

Auditory setup

McGinley lab

Equipment used or suggested by the McGinley lab.

Item	Description	Count	Source	Cost (\$)
PXIe-1082 chassis	Holds and powers boards	1	eBay / used	2,650.00
PXIe-6363 signal board	Records timing markers	1	eBay / used	2,495.36
PXI-4461 audio module, hybrid 186900-11L	Generates the sound waveform	1	eBay / used	699.99
PXIe-8360 module	Links chassis to computer	1	eBay / used	493.20
PCIe-8361 computer card	Computer connection to chassis	1	eBay / listed new	197.00
779500-03 cable, 3 metres	Links the two connection cards	1	eBay / listed new	89.85
SHC68-68-EPM, 192061C-02	Signal board to connection panel	1	eBay / used	89.99
BNC-2090A connection panel	Cable sockets for signals	1	eBay / used	613.77
ED1 driver + enclosure/power supply	Drives the ES1 speaker	1	eBay / used	209.06
ES1 speaker + matching cable	Plays tones from 4 kilohertz	1	Estimate	500.00
Hosa BNC-58-103 cable	Sound output to speaker driver	1	Pro Audio / new	13.53
Added microphone and calibration kit	Measures and calibrates sound	1	See Compact	2,517.22
Estimated parts total				10,568.97

SONIC

Setup used in the SONIC paper.

Item	Description	Count	Source	Cost (\$)
USB-6361, 782255-01	Sound output and signal recording	1	Newark / new	3,744.42
TimeMachines TM2500C	Satellite timing reference	1	New / backorder	799.99
MF1-M speaker kit	Plays the SONIC tones	1	Estimate	1,500.00
SA1 speaker amplifier	Drives the speaker	1	Estimate	2,000.00
ZB1PS powered enclosure	Houses and powers SA1	1	Estimate	750.00
781513-01 power supply	Powers the USB-6361	1	Newark / new	123.42
780534-01 locking cable	Computer to sound device	1	DigiKey / new	69.75
763830-01 power cord	Wall outlet to power supply	1	Newark / new	51.00
781233-02 cable noise filter	Reduces power cable interference	1	DigiKey / new	38.25
Hosa BNC-58-103 cable	Sound output to speaker driver	1	Pro Audio / new	13.53
Added microphone and calibration kit	Measures and calibrates sound	1	See Compact	2,517.22
Estimated parts total				11,607.58

Basic

Based on my research, this is an option for a speaker and microphone setup. We could borrow the calibrator.

Item	Description	Count	Source	Cost (\$)
Behringer UMC204HD interface	Sound playback and recording	1	eBay / new	97.98
JBL 305P MkII powered speaker	Plays tones; built-in amplifier	1	Sweetwater / new	99.00
Dayton EMM-6 microphone	Measures sound at the ear	1	eBay / new	66.93
Hosa HMIC-010 microphone cable	Microphone to sound interface	1	Sweetwater / new	17.99
Hosa HSS-005 cable, \$14.95 each	Speaker and audio loopback	2	Sweetwater / new	29.90
Gator RI-MICTP-FBM stand	Positions the microphone	1	Sweetwater / new	34.99
REED R8090 calibrator	Checks level at 1 kilohertz	1	DigiKey / new	219.00
Parts total				565.79

Compact

A simpler setup for the paper's full frequency range, recommended by AI.

Item	Description	Count	Source	Cost (\$)
USB-6361, 781442-01	Sound output and signal recording	1	Newark / new	3,030.42
MF2-M speaker kit	Plays the full SONIC tone range	1	Estimate	1,500.00
SA1 speaker amplifier	Drives the speaker	1	Estimate	2,000.00
ZB1PS powered enclosure	Houses and powers SA1	1	Estimate	750.00
Pomona EM4970-36# output lead	Sound output to speaker amplifier	1	DigiKey / new	23.68
Hosa PHX-106F-BULK lead	Microphone input; custom wiring	1	B&H / new	13.95
Yageo 100,000-ohm resistor, \$0.10 each	Input ground reference; proposed	2	DigiKey / new	0.20
763830-01 power cord	Wall outlet to power supply	1	Newark / new	51.00
<i>Microphone and calibration kit</i>				
Earthworks M50 microphone	Measures high-frequency sound	1	eBay / new	1,299.00
Sound Devices MM-1 amplifier	Powers microphone; boosts signal	1	B&H / new	945.00
REED R8090 calibrator	Checks level at 1 kilohertz	1	DigiKey / new	219.00
Hosa HMIC-010 microphone cable	Microphone to amplifier	1	Sweetwater / new	17.99
Gator RI-MICTP-FBM stand	Positions the microphone	1	Sweetwater / new	34.99
Panasonic LR6XWA/2SB battery, \$0.62 each	Powers the MM-1	2	DigiKey / new	1.24
Estimated parts total				9,886.47